

Mathematics

Year 6

Cambridge Primary
Student Manual



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British English edition.

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INTRODUCTION

How this book works

In this first unit you will handle numbers from a million right down to hundredths, compare decimals with confidence, and use place value to crack digit puzzles.

How to help

- **Work together.** Read the page aloud, then let your child lead.
- **Talk lots.** The questions start conversations, they do not test.
- **Keep it fun.** Short, happy sessions beat long, tired ones.

A note for grown-ups

This book follows the spirit of an international primary programme, interpreted afresh for Prime School. Every unit grows new skills out of a topic children already care about, and finishes with something to make, draw or write.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
In this unit, you will	The goals for the unit.
Let us start	A quick warm-up to wake up what you know.
Listening and speaking	Talk and listen about the topic.
Vocabulary	Learn new words, with pictures.
Remember!	A reminder of something you already learned.
Word book	Write down new words to keep.
Thinking time	A question to think and talk about.
I can	Statements to check your progress.

How to work

Take your time and look at the pictures: they are there to help you. When you see a new word, say it out loud. In Talking time there are no wrong answers, only good ideas.

GETTING READY

Before Unit 1

You do not need much: a pencil, some crayons, a notebook, and someone to work with.

Your toolkit

- Your eyes, for looking carefully at the pictures.
- Your voice, for reading out loud and sharing ideas.
- Your notebook, for writing and drawing what you find.

A little warm-up

- 1 What do you already know about this topic?
- 2 What would you most like to find out?
- 3 Who could you ask if you got stuck?

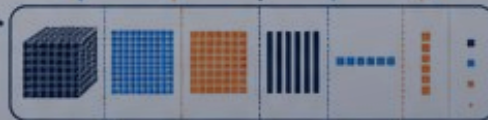
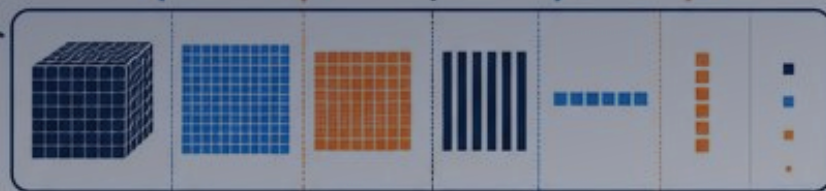
These are exactly the things Unit 1 will help you with. When you can answer them, you are ready to begin.

UNIT 1

1

Place value and decimals

From millions down to hundredths



© IN THIS UNIT, YOU WILL

- ✓ Read and write numbers to a million and beyond.
- ✓ Say what each digit is worth, including decimals.
- ✓ Compare and order decimal numbers.
- ✓ Use place value to solve digit puzzles.

◆ LET US START

Write these numbers in figures, then read them aloud to a partner: one million; five hundred thousand and five; four hundred and three thousand. Which was hardest to write, and why?

UNIT 1 • PART 1

Numbers to a million

- 1 Write these numbers in words: a. 345 678 b. 537 914 c. 303 033
- 2 Write in figures:
 - a. $200\,000 + 6000 + 300 + 2$
 - b. $900\,000 + 90\,000 + 900 + 9$
- 3 What value does the digit 7 have in each number?
 - a. 670 346 b. 702 138 c. 234 567

UNIT 1 • PART 2

Decimal place value

After the decimal point come **tenths** and then **hundredths**. In 35.46 the 3 is worth 30, the 5 is worth 5, the 4 is worth four tenths and the 6 is worth six hundredths.

Digit puzzles

Noura has these six digit cards: 9, 3, 6, 8, 1, 9.

- 1 What is the largest even number she can make using all six cards?
- 2 What is the smallest odd number she can make using all six cards?
- 3 Explain how you decided where to put each digit.

◆ THINKING TIME

Why does the last digit decide whether a number is even or odd, no matter how big the number is?

☑ SUMMARY CHECKLIST

- ✓ I can read and write numbers to a million.
- ✓ I can say what each digit is worth, including decimals.
- ✓ I can compare and order decimals.
- ✓ I can use place value to solve digit puzzles.

The digit challenge

★ FINAL PROJECT

The digit challenge

Design a digit puzzle of your own, like the one in Part 3, and write a full solution explaining the reasoning. Then swap with another group and solve theirs.

What to hand in

- Your puzzle, with clear rules.
- Your own full worked solution.
- Your written solution to another group's puzzle.

How it will be judged

Criterion	Weight	--- ---	Quality of the puzzle	30%	Correctness of your solution	35%	
Quality of reasoning	25%	Solving another group's puzzle	10%				

UNIT 1 • CHECK YOUR PROGRESS

- 1 Write 345 678 in words.
- 2 Write in figures: $200\,000 + 6000 + 300 + 2$
- 3 What is each digit worth in 158.35?
- 4 Put these in order, smallest first: 3.5, 3.05, 3.55, 3.15.
- 5 Using the digits 9, 3, 6, 8, 1, 9, what is the largest even number?
- 6 Write a decimal between 4.2 and 4.3.

UNIT 1 • EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
read and write numbers to a million	☐	☐	☐
say what each digit is worth, including decimals	☐	☐	☐
compare and order decimals	☐	☐	☐
use place value to solve digit puzzles	☐	☐	☐

UNIT 1 • WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|--|--|
| ☐ Read and write numbers to a million. | ☐ Compare and order decimals. |
| ☐ Say what each digit is worth, including decimals. | ☐ Use place value to solve digit puzzles. |

GLOSSARY

Unit 1

decimal point: the dot separating whole numbers from parts.

tenths: the first decimal place.

hundredths: the second decimal place.

place value: what a digit is worth because of its position.

SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

- 1 **Decimal place value.** The teaching sequence follows established primary-number practice.

From millions down to hundredths.

This Year 6 Student Book completes the number system. You will read and write numbers to a million, master decimal place value, order decimals, and solve digit puzzles by reasoning.

Written for Prime School pupils, in clear British English, with every answer checked by computer.

MILLIONS

DECIMALS

ORDERING

PUZZLES